

Alessio Bottini

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Education:

01/11/2019 – 06/11/2023 Ph.D in Mathematics, joint (Cotutelle) program between Université Paris–Saclay & Università di Roma Tor Vergata.
Advisors: Emanuele Macrì & Antonio Rapagnetta.
Thesis: *Stable sheaves on hyper-Kähler manifolds*
01/09/2017 – 18/09/2019 Master's degree in Mathematics, Sapienza University of Rome
01/09/2014 – 18/09/2017 Bachelor's degree in Mathematics, Sapienza University of Rome

Employment:

Present **Chapman Fellow**, Imperial College London
01/09/2025 – 31/08/2026 **Postdoctoral researcher**, Universität Bonn
01/09/2024 – 31/08/2025 **Postdoctoral researcher**, MPIM Bonn
01/09/2023 – 31/08/2024 **Postdoctoral researcher**, Universität Bonn

Research Papers:

- (8) The tangent bundle of a generalized Kummer fourfold is obstructed, *arXiv:2608.24860*.
- (7) Semi-rigid stable sheaves: a criterion and examples, with Riccardo Carini, *arXiv:2603.09477*.
- (6) The period-index problem for hyperkähler varieties: Lower and upper bounds, with Daniel Huybrechts *arXiv:2512.15131*
- (5) Benchmark question author for *IMProofBench: Benchmarking AI on Research-Level Mathematical Proof Generation*, *arXiv:2509.26076* (2025).
- (4) Derived categories of Fano varieties of lines, with Daniel Huybrechts, *arXiv:2501.03534*.
- (3) O'Grady's tenfolds from stable bundles on hyper-Kähler fourfolds, *arXiv:2411.18528*.
- (2) Towards a modular construction of OG10, *Compositio Mathematica*, 160(10):2496–2529, 2024.
- (1) Stable sheaves on K3 surfaces via wall-crossing, *Kyoto Journal of Mathematics*, 64(2):459–499, 2024.

Expository Writing:

- (3) Hyper-Kähler varieties: Lagrangian fibrations, atomic sheaves, and categories, with Emanuele Macrì and Paolo Stellari, *arXiv:2603.23033*.
- (2) A modular construction of OG10, *Oberwolfach Reports*, 2024.
- (1) The Looijenga–Lunts–Verbitsky algebra and Verbitsky's Theorem, *Milan Journal of Mathematics*, 90(2):403–416, 2022.

Honors and Awards:

- *Chapman Fellowship* at Imperial College London.
- *French Qualification* to be a Maître de conférences, obtained in 2025.
- *Fellowship* from the Hausdorff Institute for Mathematics in Bonn for the program "Derived categories, Hodge Theory, and Chow Groups" between September and December 2023. I later declined this fellowship, as I obtained a year-long contract with the University of Bonn.
- *INDAM Fellowship for research* in 2017, roughly €2000. Eight master students are awarded this fellowship based on a nation-wide competition. It also comes with a small grant (€1000) to cover research travel.

Research seminars:

- 2026 - Invited talk, Columbia University, 6 November (scheduled)
- Invited talk, Princeton University, 3 November (scheduled)
- Invited talk, University of Cambridge, 14 October (scheduled)

- Invited talk, Imperial College London, 2 October (scheduled)
- *The Period-Index problem for hyper-Kähler varieties*, University of Strasbourg
- *The Period-Index problem for hyper-Kähler varieties*, Humboldt University of Berlin
- *Sheaves on hyper-Kähler manifolds*, Oberwolfach
- 2025 - *Moduli spaces of sheaves on hyper-Kähler manifolds*, NCTS Algebraic Geometry seminar at the National Taiwan University (online talk)
- *Moduli of sheaves on hyper-Kähler manifolds*, Algebraic Geometry seminar at the University of Bielefeld
- *Moduli of sheaves on hyper-Kähler manifolds*. Oberseminar (colloquium-style talk for a general audience) at the Max Planck Institute, Bonn
- *Moduli spaces of sheaves on hyper-Kähler manifolds*, Algebraic Geometry seminar at the University of Glasgow
- *A modular construction of OG10*, HK days in Sheffield
- 2024 - *The derived category of the surface of lines on a cubic fourfold*, Hyperkähler varieties and related topics II - Rome
- *A modular construction of OG10*, Oberwolfach
- *A modular construction of OG10*, Seminario di Algebra e Geometria Università di Genova, Genova
- *Towards a modular construction of OG10*, Algebraic geometry seminar Columbia University, New York
- 2023 - *Una costruzione modulare di OG10*, Giornate di Geometria Algebrica e Argomenti Correlati XVI-Cetraro
- *Towards a modular construction of OG10*, MAGIC seminar Imperial College, London
- *Towards a modular construction of OG10*, HyperK "Topics in hyperkähler geometry" January 2023 special event - Jussieu, Paris
- 2022 - *Towards a modular construction of OG10*, Seminari di Natale Università La Statale, Milano
- *Moduli spaces of stable sheaves via wall-crossing*, Algebraic geometry seminar University of Lille
- 2021 - *Moduli spaces of stable sheaves on K3 surfaces via wall-crossing*, Algebraic geometry seminar University of Rome "Roma Tre"
- *Moduli spaces of stable sheaves on K3 surfaces via wall-crossing*, Moduli spaces and hyper-Kähler varieties in Rome, Sapienza University of Rome:
- *Moduli spaces of stable sheaves on K3 surfaces via wall-crossing*, KCL/UCL Junior Geometry Seminar- online seminar
- *A new proof of Yoshioka's Theorem on moduli spaces of sheaves*, Algebraic geometry seminar - University of Milan

Working Seminars:

- 2026 - *IHS with large second Betti numbers (after Liu-Liu-Xu)*, Bonn
- *The Period-Index conjecture for abelian varieties*
- 2025 - *The period-index problem for abelian threefolds*, in the reading seminar 'Brauer groups and the period-index problem', Bonn
- *The semiregularity map for twisted sheaves*, in the seminar 'The Hodge Conjecture for Abelian Fourfolds, after Markman', Paris
- *Characteristic cycles and the monodromy of the Gauss map*, in the seminar 'Arithmetic finiteness of very irregular varieties', Bonn
- 2024 - *Odaka's Theorems*, Lecture for the reading course 'K-stability', Bonn
- 2023 - *Derived categories of hyper-Kähler manifolds*, Junior Geometry Seminar Imperial College, London
- 2022 - *Derived categories of hyper-Kähler manifolds*, Junior talk at: Hyperkähler varieties and related topics Sapienza, Rome
- *Deformations of affine symplectic varieties*, Lecture for the reading course 'Dualité symplectique', Orsay
- *Vector bundles on hyper-Kähler varieties*, HyperK "Topics in hyperkähler geometry", Jussieu, Paris
- 2021 - *The Looijenga-Lunts algebra and Verbitsky's theorem*, Joint working group Bonn-Paris: "Topics in hyperkähler geometry" online seminar
- 2019 - *Three lectures for the reading course Reading course on stability conditions* at Sapienza University of Rome

2017 - *Kan extensions and derived functors*, Junior seminars of Algebra, Geometry and Categories University of Rome 'La Sapienza'

Academic Visits:

- Università di Roma Tor Vergata, Roma, 6-10 May 2024. Hosted by Antonio Rapagnetta.
- Columbia University, New York, 18-22 March 2024. Hosted by Giulia Saccà.
- Università La Statale, Milano, 5-9 February 2024. Hosted by Paolo Stellari.
- Imperial College, London, 6-10 February 2023. Hosted by Richard Thomas.
- MPI Leipzig, 17-26 August 2022. Hosted by Angel Rios Ortiz.

Review:

I reviewed papers for: EpiGA, Crelle's Journal (x2), Annali della Scuola Normale Superiore di Pisa, Proceedings of the London Mathematical Society, Journal of Differential Geometry, Canadian Journal of Mathematics, Mathematische Zeitschrift.

Teaching experience:

- WS 2025/2026 V5A4, Selected topics in Algebraic Geometry: Hyper-Kähler manifolds, Universität Bonn;
- Ph.D. courses:
 - Sheaves on hyper-Kähler manifolds, Pragmatic school, University of Catania, 2025.
 - Sheaves on hyper-Kähler manifolds, University of Rome "Tor Vergata", 2024.
 - Sheaves on hyper-Kähler manifolds, University of Milano Statale, 2024.
 - Sheaves on hyper-Kähler manifolds and deformation theory, in JTP "Derived categories, Hodge theory and Chow groups" - HIM, Bonn 2023
 - Examples of atomic bundles, Exercise session Summer School in Algebraic Geometry: *Derived Categories, Stability Conditions, and Moduli Spaces*, Haifa